DataSynth

Enterprise Synthetic Data Platform

High-Performance Generation for Accounting, Audit & ML

Rust core • Python wrapper • REST/gRPC API • Desktop UI

Executive Overview

Version 0.3.0

Deterministic, privacy-preserving synthetic data generation
for enterprise finance, compliance testing, and machine learning.

Built with Rust & Python • 16 modular crates • 100 K+ entries/sec

Open-source (Apache-2.0) • Commercial license available

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1 Executive Summary

DataSynth is a high-performance synthetic data platform purpose-built for enterprise accounting, audit analytics, and machine learning. Written in Rust for maximum throughput and memory safety — with a full Python wrapper for data-science workflows — it generates realistic, internally coherent financial data that satisfies the statistical and structural properties of real-world enterprise resource planning (ERP) systems.

Why DataSynth?

- **No real data required** — eliminates privacy, regulatory, and procurement barriers to analytics development.
- **Full auditability** — deterministic, seeded generation means every dataset is perfectly reproducible.
- **ML-ready from day one** — ground-truth labels for fraud, anomalies, and data quality ship alongside every record.
- **Domain depth** — covers the full accounting lifecycle: journal entries, document flows, subledgers, FX, intercompany, period close, and banking/AML.
- **Empirically grounded** — statistical distributions for journal entry line items, amount patterns, and temporal volumes are calibrated against empirical research conducted on real-world enterprise datasets, ensuring synthetic output mirrors the structural properties of production ERP data.

1.1 At a Glance

100 K+ entries/sec (single thread)	**50+** ACFE-aligned fraud schemes	**10** industry presets
16 modular Rust crates	**40+** red flag patterns	**4** privacy levels

2 Platform Capabilities

2.1 Enterprise Accounting Simulation

DataSynth generates a complete, internally consistent accounting universe:

Domain	Capabilities
Journal Entries	Balanced debits/credits, Benford-compliant amounts, configurable line-item distributions, SAP ACDOCA format export.
Master Data	Vendors, customers, materials, fixed assets, employees with hierarchies, payment terms, credit ratings, and intercompany flags.
Document Flows	Full Procure-to-Pay (PO → GR → Invoice → Payment) and Order-to-Cash (SO → Delivery → Invoice → Receipt) with three-way match validation.
Intercompany	Matched IC journal entry pairs, transfer pricing (Cost-Plus, Resale-Minus, CUP), and consolidation elimination entries.
Subledgers	AR/AP open items and aging, fixed asset register with depreciation schedules, inventory positions and movements, GL-to-subledger reconciliation.
FX & Translation	Ornstein–Uhlenbeck exchange rate process, multi-currency trial balance translation, currency translation adjustment entries.
Period Close	Month-end accruals, depreciation runs, year-end closing entries, fiscal period status tracking.

2.2 Banking, KYC & AML

A dedicated banking module generates realistic transaction data for anti-money-laundering testing:

- **Customer personas**: Retail, Business, Trust profiles with full KYC envelopes (declared turnover, source of funds, geographic exposure, cash intensity).
- **AML typologies**: Structuring, funnel accounts, layering schemes, money mule networks, round-tripping, and adversarial spoofing for robustness testing.
- **Ground-truth labels**: Entity-level risk classifications, transaction-level labels, and investigation narratives.

2.3 Audit Simulation

Generates ISA-compliant audit artifacts:

- Engagement metadata with materiality thresholds (ISA 320).
- Workpapers per ISA 230, evidence per ISA 500.
- Risk assessments (ISA 315/330), findings (ISA 265), and professional judgment documentation (ISA 200).

2.4 COSO 2013 Internal Control Framework

Full integration with the COSO Internal Control-Integrated Framework:

- **5 Components**: Control Environment, Risk Assessment, Control Activities, Information & Communication, Monitoring Activities.
- **17 Principles**: Complete principle coverage with control mappings.
- **Control Scopes**: Entity-level, transaction-level, IT general, and IT application controls.
- **Maturity Levels**: 6-level model (Non-Existent through Optimized).
- **Export**: `coso_control_mapping.csv` with principle-level granularity.

2.5 Accounting & Audit Standards

Comprehensive standards support via the `datasynth-standards` crate:

Category	Standards Supported
Accounting (US GAAP)	ASC 606 (Revenue), ASC 842 (Leases), ASC 820 (Fair Value), ASC 360 (Impairment)
Accounting (IFRS)	IFRS 15 (Revenue), IFRS 16 (Leases), IFRS 13 (Fair Value), IAS 36 (Impairment)
Audit Standards	34 ISA standards (ISA 200–720), 19+ PCAOB standards with ISA mapping
Regulatory	SOX Section 302 (Certifications), SOX Section 404 (ICFR Assessment), Deficiency Matrix (MW/SD classification)

Supports dual reporting mode (US GAAP + IFRS) with automatic framework difference reconciliation.

2.6 Process Mining (OCEL 2.0 & XES 2.0)

Object-Centric Event Logs track many-to-many relationships between business objects (orders, invoices, payments) and activities — enabling conformance checking and process variant analysis.

- **OCEL 2.0**: JSON and XML export with objects, events, and object relationships.
- **XES 2.0**: IEEE standard format for ProM, Celonis, Disco, and pm4py.
- **Reference Models**: Canonical P2P (9 activities), O2C (10 activities), and R2R (11 activities) process models for conformance checking.
- **17 R2R/A2R Activities**: GL posting, FX revaluation, period close, depreciation, consolidation.
- **Deterministic UUIDs**: FNV-1a hash-based IDs with type discriminators for reproducible event logs.

3 Machine Learning & Analytics

3.1 ACFE-Aligned Fraud Taxonomy

DataSynth implements the ACFE (Association of Certified Fraud Examiners) "Report to the Nations" taxonomy with calibrated statistics:

ACFE Category	Prevalence	Scheme Types
Asset Misappropriation	86%	20+ cash schemes: skimming, larceny, shell company, ghost employee, billing fraud, expense reimbursement
Corruption	33%	Conflicts of interest, bribery, kickbacks, bid rigging, economic extortion
Financial Statement	10%	Revenue manipulation, expense timing, concealed liabilities, improper disclosures

Calibration: $117k median loss, 12-month median duration, detection method distribution (42% tips, 16% internal audit, 12% management review).

3.2 Advanced Fraud Modeling

- **Collusion & Conspiracy**: 9 ring types (EmployeePair, DepartmentRing, EmployeeVendor, VendorRing), 6 conspirator roles, defection modeling based on detection risk.
- **Management Override**: Executive-level fraud with override techniques, Fraud Triangle modeling (Pressure, Opportunity, Rationalization), concealment strategies.
- **Red Flag Patterns**: 40+ patterns with Bayesian probabilities, 6 categories (Vendor, Transaction, Timing, Approval, Document, Behavioral), realistic false positive rates.
- **Industry-Specific**: Manufacturing (yield manipulation, phantom production), Retail (sweethearting, refund fraud), Healthcare (upcoding, unbundling, phantom billing).

Every injected anomaly carries a `LabeledAnomaly` record with full metadata, enabling supervised and semi-supervised learning pipelines without manual labeling effort.

3.3 Data Quality Variations

Realistic data imperfections for training data-quality ML models:

- **Missing values**: MCAR, MAR, MNAR, and systematic patterns.
- **Format variations**: Date, amount, and identifier format diversity across regional conventions.
- **Duplicates**: Exact, near, and fuzzy duplicates.
- **Typos**: Keyboard-aware substitution, OCR errors, homophones.
- **Encoding issues**: Mojibake, BOM artifacts, HTML entity corruption.

3.4 Enterprise Interconnectivity

Realistic enterprise network modeling with configurable relationships:

- **Multi-Tier Vendor Networks**: 3-tier supply chain (Tier1/Tier2/Tier3), vendor clusters (ReliableStrategic 20%, StandardOperational 50%, Transactional 25%, Problematic 5%), lifecycle stages, quality scores, dependency analysis.
- **Customer Value Segmentation**: Enterprise (40% revenue/5% customers), MidMarket (35%/20%), SMB (20%/50%), Consumer (5%/25%), lifecycle stages, referral networks, corporate hierarchies.
- **Entity Relationship Graph**: 16 entity types, 26 relationship types, composite strength calculation from volume, count, duration, recency, and mutual connections.
- **Cross-Process Links**: P2P↔O2C linkage via inventory (GoodsReceipt→Delivery), intercompany bilateral links.

3.5 Pattern & Process Drift

Comprehensive drift modeling for realistic temporal evolution:

- **Organizational Events**: Acquisition (1.35x volume, 5% integration errors), divestiture, reorganization, leadership change, workforce reduction, merger.
- **Process Evolution**: Approval workflow changes, S-curve automation rollout, policy changes, control enhancements.
- **Technology Transitions**: ERP migration phases with error rate and processing time multipliers, duplicate/missing/format issues.
- **Behavioral Drift**: Vendor payment term extensions, customer delays during downturns, employee learning curves, year-end intensity.
- **Market Drift**: Economic cycles (sinusoidal, asymmetric, mean-reverting), industry-specific cycles, commodity drift, price shocks.
- **Ground Truth**: `LabeledDriftEvent` with event type, magnitude, and detection difficulty (Easy/Medium/Hard) for drift detection ML.

3.6 Graph & Network Export

Supported Graph Formats

PyTorch Geometric	`.pt` files with node features, edge index, edge attributes, labels, and train/val/test masks.
Neo4j	CSV node/edge files with Cypher import scripts.
DGL	Deep Graph Library format for GNN training.

Computed features include temporal signals (weekday, period-end flags), amount signals (log-amount, Benford probability, round-number flag), structural signals (line count, unique accounts), and one-hot categorical encodings.

4 Privacy & Fingerprinting

4.1 Privacy-Preserving Fingerprint Extraction

DataSynth can extract a statistical *fingerprint* from real data and synthesize new data that matches its properties — without ever copying individual records.

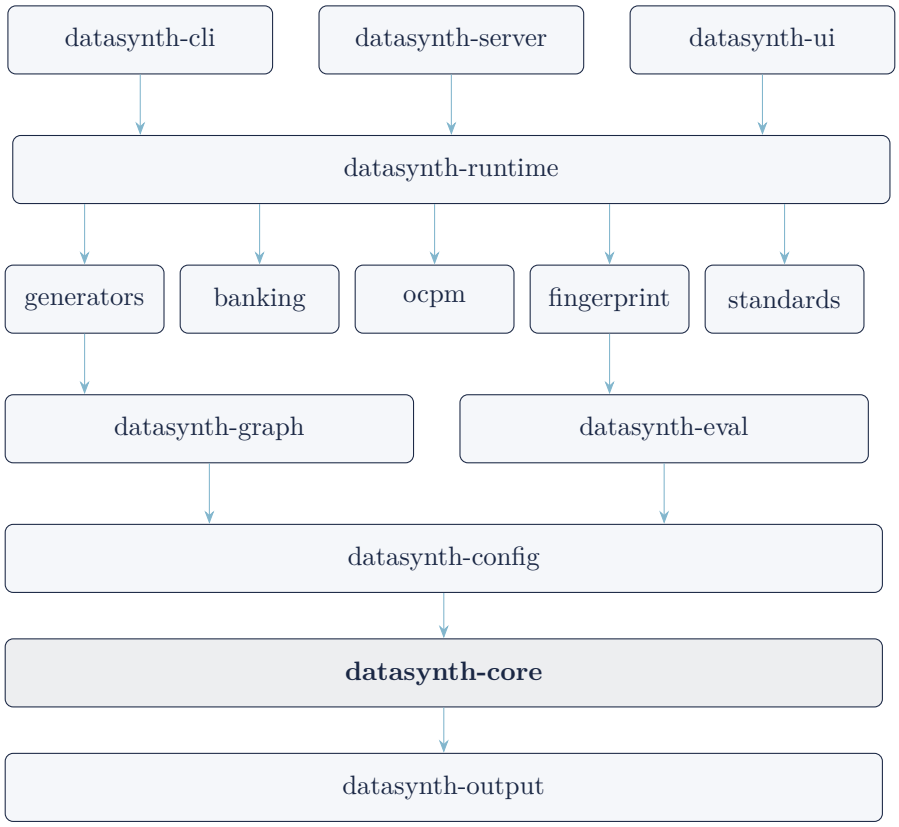
4.2 Privacy Levels

Level	Epsilon (ε)	k-Anon	Outlier %	Use Case
Minimal	5.0	3	99%	Low privacy, high utility
Standard	1.0	5	95%	Balanced (default)
High	0.5	10	90%	Sensitive environments
Maximum	0.1	20	85%	Maximum privacy guarantees

The privacy engine combines differential privacy (Laplace and Gaussian mechanisms), k-anonymity with rare-value suppression, and outlier winsorization. A full privacy audit trail records every decision and the cumulative epsilon budget spent.

5 Architecture

5.1 Layered Crate Architecture



5.2 Production-Grade Infrastructure

Capability	Details
Deterministic Output	ChaCha8 PRNG with configurable seed; identical seed ⇒ identical dataset.
Financial Precision	`rust_decimal` throughout; no IEEE 754 floating-point artifacts. Decimals serialized as strings.
Resource Guards	Unified CPU, memory, and disk monitoring with automatic throttling and graceful degradation (Normal → Reduced → Minimal → Emergency).
Collision-Free IDs	FNV-1a hash-based UUID factory with generator-type discriminators prevents document ID collisions across parallel generators.
Streaming Output	CSV, JSON, NDJSON, and Parquet streaming sinks with backpressure strategies (Block, DropOldest, DropNewest, Buffer).
API Layer	REST + gRPC + WebSocket with API-key authentication, sliding-window rate limiting, and configurable timeouts.
Desktop UI	Tauri + SvelteKit cross-platform application with 15+ configuration pages and real-time streaming viewer.
Python Wrapper	`datasynth-py` package with blueprints, pandas integration, and WebSocket streaming support.

6 Industry Presets

DataSynth ships with ten industry-specific configuration presets, each tuned for realistic business process weightings, chart-of-accounts structures, and regional multi-company setups.

Industry	Key Weight	Characteristics
Manufacturing	40% P2P	14 transaction types, multi-level BOM, production routings, work centers, JIT, quality framework.
Retail	50% O2C	12 transaction types, 7 store types, shrinkage modeling, loss prevention, markdown patterns.
Healthcare	15% H2R	15 transaction types, payer mix (Medicare/Medicaid/Commercial), ICD-10/CPT/DRG coding, HIPAA/Stark compliance.
Financial Services	40% R2R	Report-heavy, regulatory compliance, intercompany structures, loan loss provisions.
Technology	15% H2R	SaaS revenue recognition, R&D capitalization, deferred revenue.
Professional Svcs.	—	Time-based billing, project accounting, trust accounting.
Energy	—	Capital-intensive, long-lived assets.
Transportation	—	Fleet management, route-based costing.
Real Estate	—	Property portfolios, lease accounting.
Telecommunications	—	Subscription revenue, network assets.

Each preset supports three complexity tiers: **Small** (∼100 GL accounts), **Medium** (∼400 accounts), and **Large** (∼2 500 accounts).

7 Statistical Distribution Framework

DataSynth includes an advanced statistical distribution framework for generating empirically grounded synthetic data:

Capability	Details
Mixture Models	Gaussian and Log-Normal mixtures with weighted components ("routine", "significant", "major" transactions).
Copula Correlation	Gaussian, Clayton, Gumbel, Frank, Student-t copulas for cross-field dependencies.
Distribution Types	Pareto (heavy-tailed), Weibull (time-to-event), Beta (proportions), Zero-inflated (excess zeros).
Benford's Law	First and second-digit compliance, round number bias, threshold clustering.
Regime Changes	Economic cycles, acquisition/divestiture effects, recession modeling.
Industry Profiles	Pre-configured profiles for Retail, Manufacturing, Financial Services.

All distributions include validation tests: Benford's Law MAD, Anderson-Darling goodness-of-fit, chi-squared, and correlation matrix verification.

8 Use Cases

Primary Use Cases

- **Fraud Detection ML** — Train and validate models on labeled fraud typologies with realistic base-rate imbalance.
- **Graph Neural Networks** — Export transaction graphs in PyTorch Geometric, Neo4j, or DGL format with pre-computed features and train/val/test splits.
- **AML & KYC Testing** — Generate banking transaction data with structuring, layering, and mule patterns for compliance system validation.
- **Audit Analytics** — Produce ISA-compliant audit artifacts and anomaly-injected financial data for analytics tool development.
- **ERP Integration Testing** — Generate SAP ACDOCA-format data with full document chains for system migration and integration testing.
- **Process Mining** — OCEL 2.0 event logs for process discovery, conformance checking, and variant analysis.
- **SOX & COSO Compliance Testing** — Internal control definitions with COSO 2013 mappings, SOX 302/404 certifications, deficiency classification, and SoD conflict detection.
- **Accounting Standards Testing** — Generate revenue contracts (ASC 606/IFRS 15), lease portfolios (ASC 842/IFRS 16), fair value measurements, and impairment tests with dual-framework reconciliation.
- **Data Quality ML** — Labeled missing values, typos, duplicates, and format variations for training data-cleansing models.
- **Drift Detection ML** — Labeled drift events (organizational, process, technology, market, regulatory) with magnitude and detection difficulty for drift detection model training.
- **Network Analytics** — Multi-tier vendor networks and customer segmentation graphs with relationship strength metrics for supply chain and customer analytics.

9 Evaluation & Auto-Tuning

DataSynth includes a built-in evaluation framework that measures synthetic data quality across four dimensions:

1. **Statistical Fidelity** — KS statistic, Wasserstein distance, Benford's Law MAD, Anderson-Darling goodness-of-fit.
2. **Coherence** — Balance-sheet validation, intercompany matching, document chain integrity, standards compliance.
3. **Data Quality** — Completeness, consistency, duplicate rates, format correctness, uniqueness.

4. **ML Readiness** — Feature distributions, label quality, graph structure, train/val/test split balance.
5. **Drift Detection** — Hellinger distance, Population Stability Index (PSI), rolling window analysis, detection delay metrics.

9.1 ML Benchmarks

Pre-configured benchmarks aligned with ACFE statistics and industry verticals:

- **ACFE-Calibrated**: General fraud (1k), collusion-focused (5k), management override (2k).
- **Industry-Specific**: Manufacturing fraud (5k), retail fraud (10k), healthcare fraud (5k), technology fraud (3k), financial services fraud (5k).
- **Cost-Sensitive**: Asymmetric cost matrices for realistic fraud detection evaluation.

An **auto-tuning engine** analyses evaluation results and produces prioritized configuration patches with expected improvement estimates — closing the loop between generation and validation.

10 Getting Started

10.1 Quick Start (CLI)

```
# Generate with demo preset
datasynth-data generate -demo -output ./output

# Create an industry-specific config
datasynth-data init -industry manufacturing -complexity medium -o config.yaml

# Generate from config
datasynth-data generate -config config.yaml -output ./output
```

10.2 Quick Start (Python)

```
from datasynth_py import DataSynth
from datasynth_py.config import blueprints

config = blueprints.retail_small(companies=4, transactions=10000)
synth = DataSynth()
result = synth.generate(config=config)
```

10.3 Server Mode

```
# Start REST/gRPC server with 4 worker threads
cargo run -p datasynth-server - -port 3000 -worker-threads 4
```

Links & Resources

Repository	https://github.com/ey-asu-rnd/SyntheticData
Documentation	https://ey-asu-rnd.github.io/SyntheticData/
Crates.io	https://crates.io/crates/datasynth-core
PyPI	https://pypi.org/project/datasynth-py/

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